SQL Query Solutions

```
# Change database to Atliq Hardwares
USE gdb023;
```

 # 1. The list of markets in which the customer "Atliq Exclusive" operates its business in the APAC region

```
SELECT
	DISTINCT market
FROM
	dim_customer
WHERE
	customer = "Atliq Exclusive" AND
	region = "APAC";
```

market
India
Indonesia
Japan
Philiphines
South Korea
Australia
Newzealand
Bangladesh

8 row(s) returned in 0.094 sec / 0.000 sec

 # 2. What is the percentage of unique product increase in 2021 vs. 2020?

```
WITH FY20 AS (
	SELECT
		COUNT(DISTINCT(product_code)) AS up_20
	FROM
		fact_sales_monthly
	WHERE
		fiscal_year = 2020
), FY21 AS (
	SELECT
		COUNT(DISTINCT(product_code)) AS up_21
	FROM
		fact_sales_monthly
	WHERE
		fiscal_year = 2021
)
SELECT
	FY20.up_20 AS unique_product_2020,
	FY21.up_21 AS unique_product_2021,
	CONCAT(ROUND((FY21.up_21 - FY20.up_20) * 100/FY20.up_20, 2), '%')AS percentage_chg
FROM
	FY20, FY21;
```

unique_product_2020	unique_product_2021	percentage_chg
245	334	36.33%

1 row(s) returned in 1.906 sec / 0.000 sec

 # 3. Provide a report with all the unique product counts for each segment and sort them in descending order of product counts.

```
SELECT
        segment,
        COUNT(product) AS product_count
FROM
        dim_product
GROUP BY
        segment
ORDER  BY
        product_count DESC;
```

segment	product_count
Notebook	129
Accessories	116
Peripherals	84
Desktop	32
Storage	27
Networking	9

6 row(s) returned in 0.079 sec / 0.000 sec

 # 4. Follow-up: Which segment had the most increase in unique products in 2021 vs 2020?

```
WITH FY20 AS (
                SELECT
                        segment,
                        COUNT(DISTINCT(s.product_code)) AS seg20
                FROM
                        fact_sales_monthly s
```

```
        JOIN
            dim_product p
        ON
            s.product_code = p.product_code
        WHERE
            fiscal_year = 2020
        GROUP BY
            p.segment
),  FY21 AS (
        SELECT
            segment,
            COUNT(DISTINCT(s.product_code)) AS seg21
        FROM
            fact_sales_monthly s
        JOIN
            dim_product p
        ON
            s.product_code = p.product_code
        WHERE
            fiscal_year = 2021
        GROUP BY
            p.segment
)
SELECT
    FY20.segment,
    seg20 AS product_count_2020,
    seg21 AS product_count_2021,
    seg21 - seg20 AS difference
FROM
    FY20
JOIN
    FY21
ON
```

```
        FY20.segment = FY21.segment
ORDER BY
        difference DESC;
```

segment	product_count_2020	product_count_2021	difference
Accessories	69	103	34
Notebook	92	108	16
Peripherals	59	75	16
Desktop	7	22	15
Storage	12	17	5
Networking	6	9	3

6 row(s) returned in 2.922 sec / 0.000 sec

 # 5. Get the products that have the highest and lowest manufacturing costs.

```
SELECT
        c.product_code,
        product,
        CONCAT(manufacturing_cost, '/unit') AS manufacturing_cost
FROM
        fact_manufacturing_cost AS c
JOIN
        dim_product p
ON
        p.product_code = c.product_code
WHERE
        c.manufacturing_cost = (
                                SELECT
                                        MAX(manufacturing_cost)
```

```
                    FROM
                        fact_manufacturing_cost
                    ) OR
        c.manufacturing_cost = (
                    SELECT
                        MIN(manufacturing_cost)
                    FROM
                        fact_manufacturing_cost
                    )
ORDER BY
        manufacturing_cost DESC;
```

product_code	product	manufacturing_cost
A6120110206	AQ HOME Allin1 Gen 2	240.5364/unit
A2118150101	AQ Master wired x1 Ms	0.8920/unit

2 row(s) returned in 0.047 sec / 0.000 sec

 # 6. Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market.

```
SELECT
        d.customer_code,
        customer,
        ROUND(AVG(pre_invoice_discount_pct) * 100, 2) AS average_discount_percentage
FROM
        fact_pre_invoice_deductions d
JOIN
        dim_customer c
ON
        d.customer_code = c.customer_code
WHERE
```

```
        market = "India" AND
        d.fiscal_year = 2021
GROUP BY
        customer,
        d.customer_code
ORDER BY
        average_discount_percentage DESC
LIMIT
        5;
```

customer_code	customer	average_discount_percentage
90002009	Flipkart	30.83
90002006	Viveks	30.38
90002003	Ezone	30.28
90002002	Croma	30.25
90002016	Amazon	29.33

5 row(s) returned in 0.016 sec / 0.000 sec

 # 7. Get the complete report of the Gross sales amount for the customer "Atliq Exclusive" for each month. This analysis helps to get an idea of low and high-performing months and take strategic decisions.

```
WITH gross_sales_table AS (
        SELECT
                date,
                s.customer_code,
                p.fiscal_year,
                gross_price * sold_quantity AS gross_sales
        FROM
                fact_gross_price p
        JOIN
```

```
                fact_sales_monthly s
        ON
                s.product_code = p.product_code AND
                s.fiscal_year = p.fiscal_year
), customer_sort AS (
        SELECT
                date,
                c.customer_code,
                gross_sales
        FROM
                gross_sales_table t
        JOIN
                dim_customer c
        ON
                t.customer_code = c.customer_code
        WHERE
                customer = "Atliq Exclusive"
)
SELECT
    MONTH(date) AS month,
    YEAR(date) AS year,
    ROUND(SUM(gross_sales) / 1000000, 2) AS gross_sales_amount_mln
FROM
    customer_sort
GROUP BY
    month, year;
```

month	year	gross_sales_amount_mln
9	2019	4.5
10	2019	5.14

11	2019	7.52
12	2019	4.83
1	2020	4.74
2	2020	4
3	2020	0.38
4	2020	0.4
5	2020	0.78
6	2020	1.7
7	2020	2.55
8	2020	2.79
9	2020	12.35
10	2020	13.22
11	2020	20.46
12	2020	12.94
1	2021	12.4
2	2021	10.13
3	2021	12.14
4	2021	7.31
5	2021	12.15
6	2021	9.82
7	2021	12.09

8	2021	7.18

24 row(s) returned in 1.750 sec / 0.000 sec

 # 8. In which quarter of 2020 got the maximum total_sold_quantity?

```
WITH quarter AS (
        SELECT
            sold_quantity,
            CASE
                WHEN
                    MONTH(date) BETWEEN 09 AND 11
                THEN
                    "Q1"
                WHEN
                    MONTH(date) IN (12, 01, 02)
                THEN
                    "Q2"
                WHEN
                    MONTH(date) BETWEEN 03 AND 05
                THEN
                    "Q3"
                WHEN
                    MONTH(date) BETWEEN 06 AND 08
                THEN
                    "Q4"
            END AS Quarter
        FROM
            fact_sales_monthly
        WHERE
            fiscal_year = 2020
)
SELECT
```

```
        Quarter,
        SUM(sold_quantity) AS total_sold_quantity
FROM
        quarter
GROUP BY
        Quarter
ORDER BY
        total_sold_quantity DESC;
```

Quarter	total_sold_quantity
Q1	7005619
Q2	6649642
Q3	2075087
Q4	5042541

4 row(s) returned in 1.265 sec / 0.000 sec

 # 9. Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution?

```
WITH gross_sale_table AS (
            SELECT
                    customer_code,
                    gross_price *sold_quantity AS gross_sales_mln
            FROM
                    fact_gross_price g
            JOIN
                    fact_sales_monthly s
            ON
                    g.product_code = s.product_code AND
                    g.fiscal_year = s.fiscal_year
```

```
        WHERE
            g.fiscal_year = 2021
), channel_table AS (
        SELECT
            channel,
            ROUND(SUM(gross_sales_mln / 1000000), 3) AS gross_sales_mln
        FROM
            gross_sale_table gt
        JOIN
            dim_customer c
        ON
            c.customer_code = gt.customer_code
        GROUP BY
            channel
), total_sum AS (
        SELECT
            SUM(gross_sales_mln) AS SUM_
        FROM
            channel_table
)
SELECT
    ct.*,
    CONCAT(ROUND(ct.gross_sales_mln * 100 / ts.SUM_, 2), '%') AS percentage
FROM
    channel_table ct, total_sum ts
ORDER BY
    percentage DESC;
```

channel	gross_sales_mln	percentage
Retailer	1219.082	73.23%
Direct	257.532	15.47%

Distributor	188.026	11.30%

3 row(s) returned in 3.328 sec / 0.000 sec

 # 10. Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021?

```
WITH product_table AS (
        SELECT
                p.division,
                s.product_code,
                p.product,
                SUM(s.sold_quantity) AS total_sold_quantity
        FROM
                fact_sales_monthly s
        JOIN
                dim_product p
        ON
                p.product_code = s.product_code
        WHERE
                s.fiscal_year  = 2021
        GROUP BY
                s.product_code,
                p.division,
                p.product
), rank_table AS (
        SELECT
                *,
                RANK() OVER(
                        PARTITION BY
                                division
                        ORDER BY
                                total_sold_quantity DESC
```

```
                                        ) AS rank_order
                    FROM
                                product_table
)
SELECT
        *
FROM
        rank_table
WHERE
        rank_order < 4;
```

division	product_code	product	total_sold_quantity	rank_order
N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1
N & S	A6818160202	AQ Pen Drive DRC	688003	2
N & S	A6819160203	AQ Pen Drive DRC	676245	3
P & A	A2319150302	AQ Gamers Ms	428498	1
P & A	A2520150501	AQ Maxima Ms	419865	2
P & A	A2520150504	AQ Maxima Ms	419471	3
PC	A4218110202	AQ Digit	17434	1
PC	A4319110306	AQ Velocity	17280	2
PC	A4218110208	AQ Digit	17275	3

9 row(s) returned in 3.266 sec / 0.000 sec